



WIRELESS WIRELESS Intrusion Detection

Wireless networks are more prone to unauthorised intrusions compared to wired networks. Are you prepared for these threats?

Wireless technology cannot be ignored any longer. Given its advantages and the conveniences that come with it, it will probably replace all current network infrastructures pretty soon. As with all new things, this technology, too, brings with it a number of potential loopholes. That's why we need to be more careful about guarding these networks from security threats.

Wireless networks are more prone to attacks when compared to wired networks. You do not have to physically connect your computer to the network to get access to a wireless network, which means that there are more chances of getting your network compromised. This makes securing a wireless network a tough job. An administrator of the network may use strong encryption and prevent the possibility of an attack to some extent, yet the use of encryption such as in WEP is not good enough and can be cracked very easily.

WPA and other new standards like WPA2 are stronger, but are still more vulnerable to attacks than traditional wired networking structures. What most common users or new users of wireless networks do not understand is, even the strongest encryption is useless if your passphrase is simple. Using passwords like '1234' or 'abcd' with strong protocols is as good as no encryption at all because it can be cracked very easily. Even with the use of stronger passwords, the advances in cracking means that if a cracker tries hard

enough, he will eventually succeed in breaking through security.

With a majority of networks (all the discussion henceforth relates to wireless networking unless mentioned otherwise) that apply simple passwords, the possibility of potential crackers entering a network is increasing day by day. This makes the detection of network intrusion a very difficult task indeed.

There are some types of attacks that are initiated from rogue access points in the vicinity of a wireless network. Such attacks have nothing to do with the encryption protocol being used. As already stated, the inherent vulnerabilities of ad-hoc networks with simple passwords, makes you a very good potential target for crackers!

An ad-hoc network is essentially one that does not have any fixed structure. The literal meaning of ad-hoc is 'for this purpose only'. What happens practically is, some nodes come into the network only to do some specific task. Devices are part of the network only for the duration of a communications session till they complete that specific task, after which they leave the network—there is no rigid structure.

Need for intrusion detection systems

Intrusion prevention measures, such as encryption and authentication, can be used in ad-hoc networks to reduce intrusions, but they do not ensure total elimination. For example,